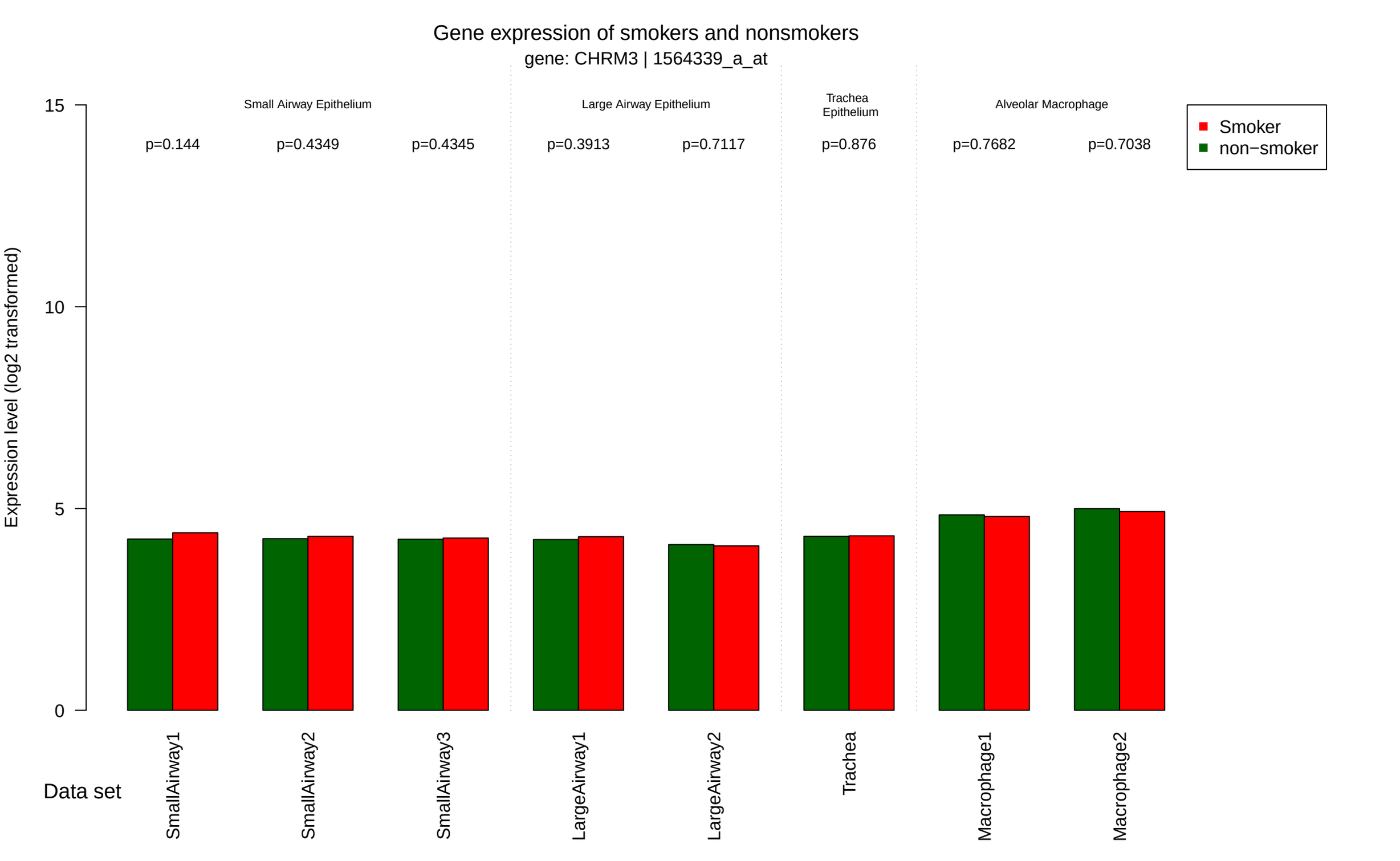
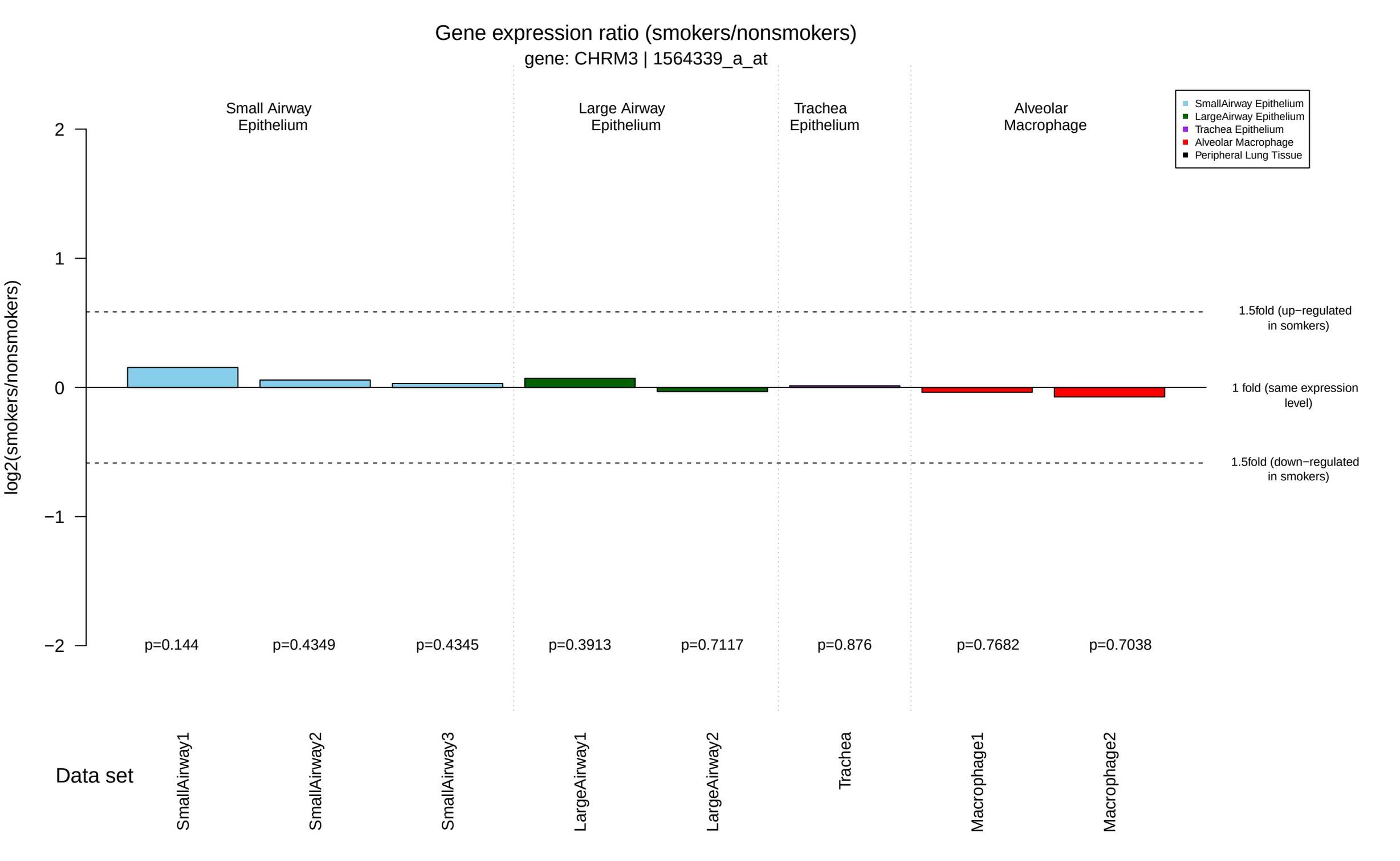
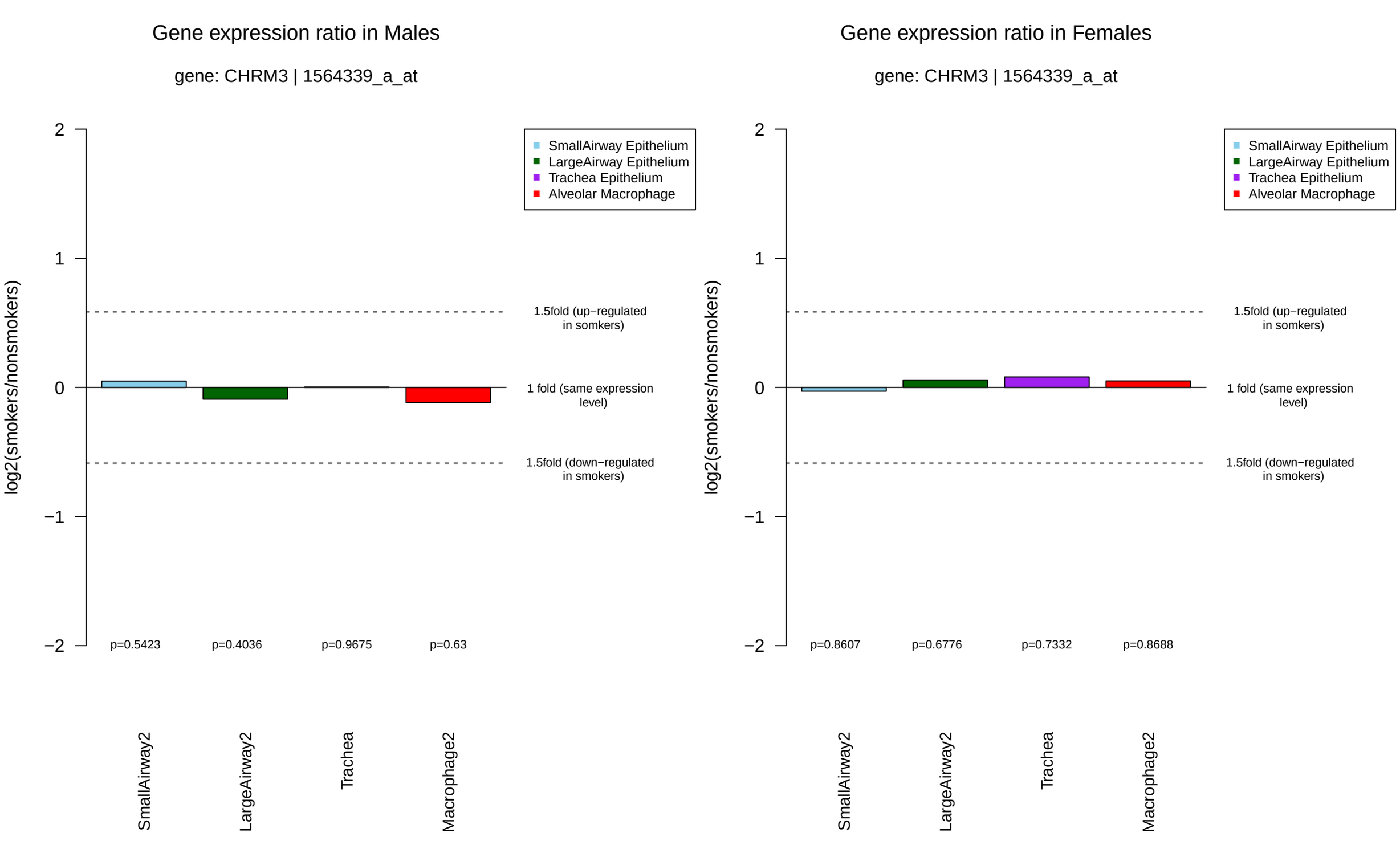
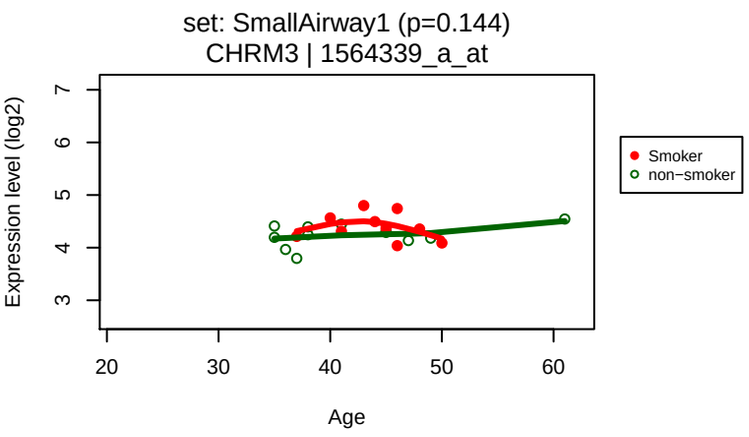
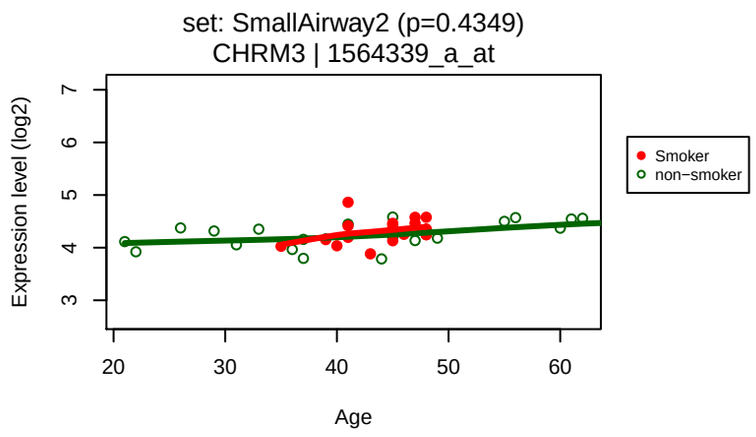
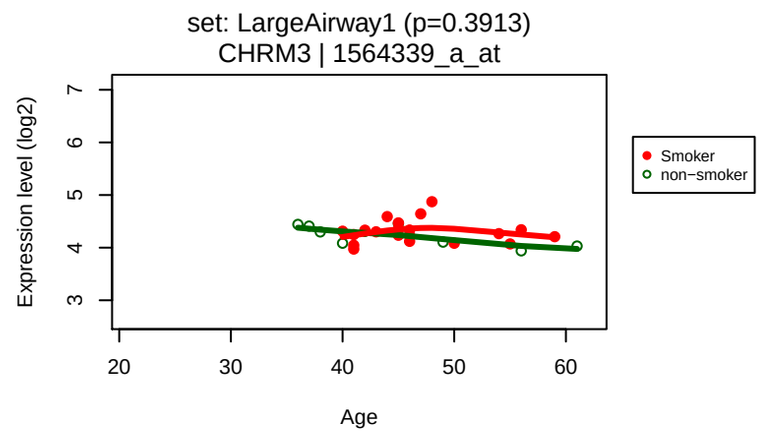

Gene expression of smokers and nonsmokers
gene: CHRM3 | 1564339_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.144
p=0.4349
p=0.4345
p=0.3913
p=0.7117
p=0.876
p=0.7682
p=0.7038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

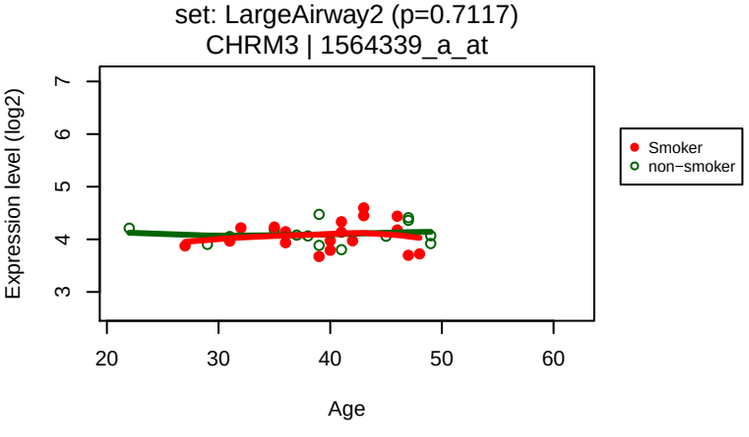

Gene expression ratio (smokers/nonsmokers)
gene: CHRM3 | 1564339_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.144
p=0.4349
p=0.4345
p=0.3913
p=0.7117
p=0.876
p=0.7682
p=0.7038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

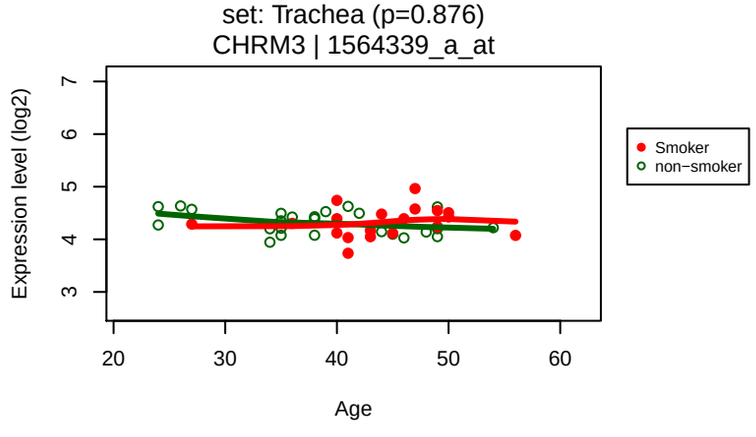

Gene expression ratio in Males
gene: CHRM3 | 1564339_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5423
p=0.4036
p=0.9675
p=0.63
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM3 | 1564339_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8607
p=0.6776
p=0.7332
p=0.8688
SmallAirway2
LargeAirway2
Trachea
Macrophage2

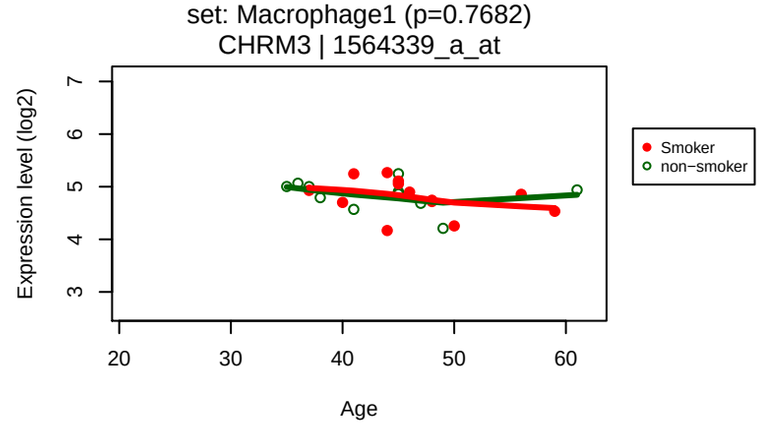

set: SmallAirway1 (p=0.144)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker

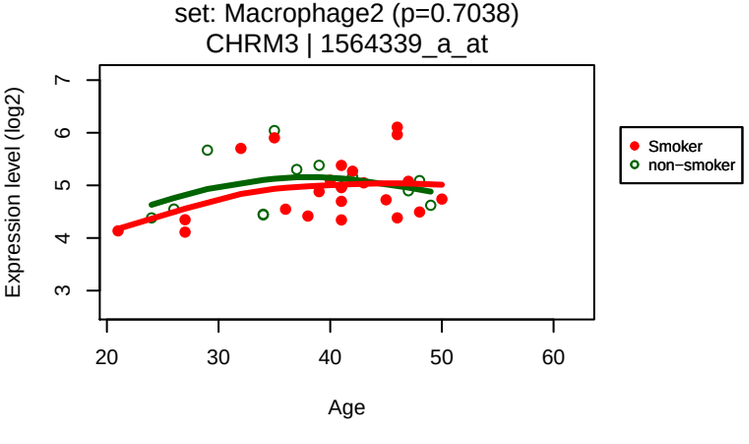

set: SmallAirway2 (p=0.4349)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker

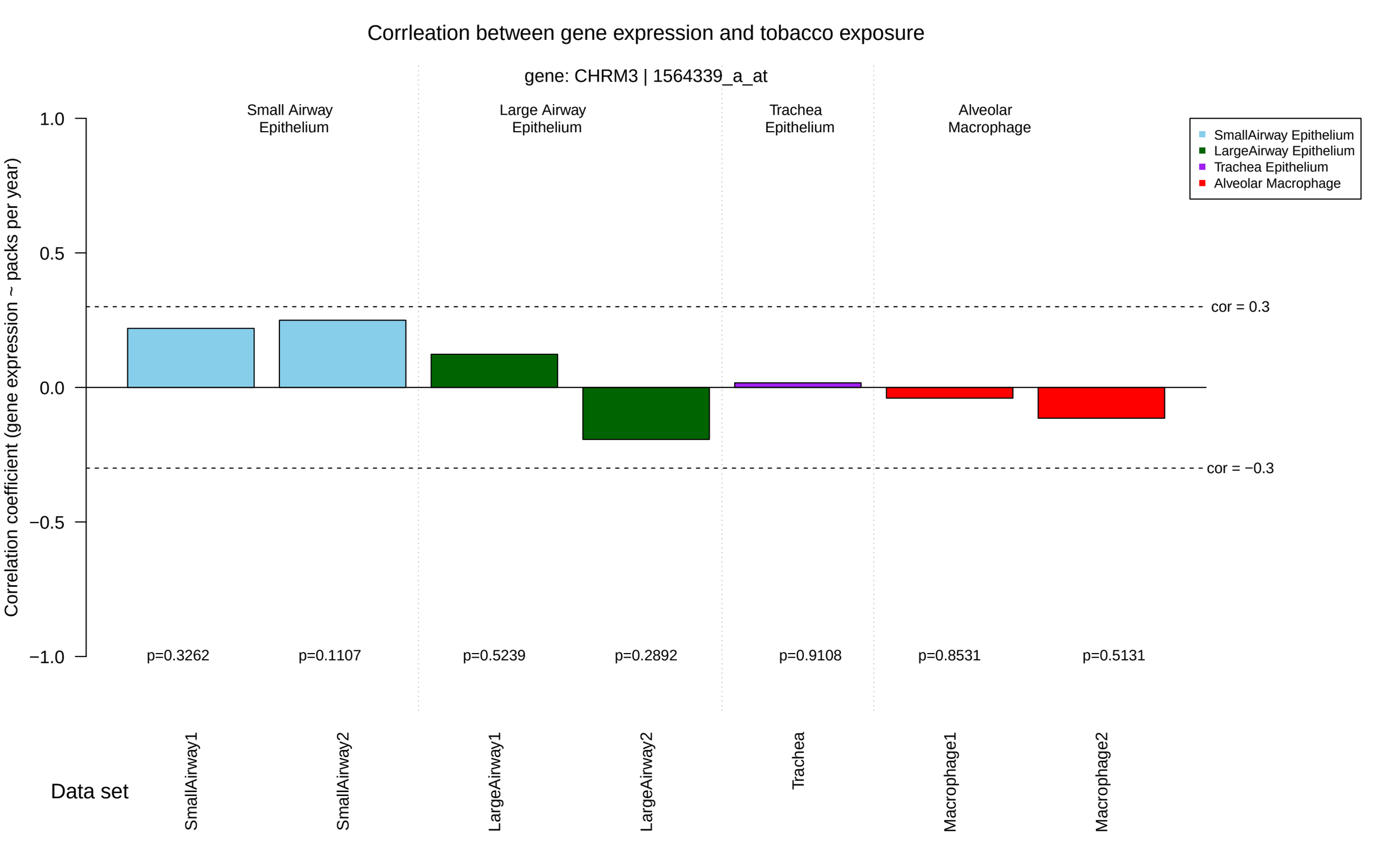

set: LargeAirway1 (p=0.3913)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7117)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.876)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7682)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7038)
CHRM3 | 1564339_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM3 | 1564339_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3262
p=0.1107
p=0.5239
p=0.2892
p=0.9108
p=0.8531
p=0.5131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2